^{34}S(n,γ) E=thermal 1985Ra15

1985Ra15: 6×10^{11} n/cm^2/s thermal neutrons at the target position were produced from the Los Alamos Omega West Reactor at a nominal 8-MW reactor power. γ rays following thermal neutron capture by a 1.09-g, 77.7% enriched ^{34}S target were detected using a 26 cm^3 coaxial Ge(Li) detector inside a 20-cm-diam by 30-cm-long NaI(Tl) annulus with FWHM=2.3 keV at 1 MeV, 5.5 keV at 6 MeV and 8.8 keV at 11 MeV. Measured σ_γ, Eγ, and Iγ for 61 γ-ray transitions. Deduced 22 levels and γ-ray branching ratios.

1972Dz13: Thermal neutrons were produced from the thermal column of the IRT reactor of the Baghdad Nuclear Research Institute. γ rays following thermal neutron capture by a 1.5-g, 97%-enriched ^{34}S target were detected using a 10-cm^3 Ge(Li) detector with FWHM=3.5 keV at 1 MeV and 8 keV at 7 MeV. Measured Eγ and Iγ for 22 γ-ray transitions. Deduced 11 levels.

1985Ke08: 5×10^{12} n/cm^2/s thermal neutrons were produced from the McMaster University Reactor. γ rays following thermal neutron capture by a natural sulfur target were detected using a 10% efficient Ge centered within a NaI(Tl) annulus with FWHM=3.2-4.4 keV at 4-7 MeV. Measured Eγ and Iγ for 13 γ-ray transitions. Deduced 9 levels.

1997Be42: Thermal neutrons were provided from the reactor BR1 of the Studiecentrum voor kernenergie in Mol, Belgium. γ rays following thermal neutron capture by a 313.6-mg, 93.26%-enriched ^{34}S and 5.933%-enriched ^{36}S target were detected using a Ge detector. Measured σ_γ, Eγ, and Iγ for 11 γ-ray transitions. Deduced 8 levels. Compared with direct-capture (DC) model calculations.

All 55 Eγ values from 2007ChZX: database of prompt gamma activation analysis (PGAA) from neutron capture are in agreement with the data below. Others: 2000Re01, 2001ZhZZ.

^{35}S Levels

E(level)[†]	Comments
0	
1572.373 *7*	
1991.28 *4*	
2347.782 *8*	C^{2}S=0.48 for 3/2$^-$ from 1997Be42.
2717.05 *9*	
2938.63 *5*	
3558.082 *27*	
3801.955 *15*	C^{2}S=0.08 for 3/2$^-$ from 1997Be42.
4105.6 *8*	
4189.243 *33*	C^{2}S=0.15 for 1/2$^-$ from 1997Be42.
4477.62 *7*	
4903.374 *15*	C^{2}S=0.49 for 1/2$^-$ from 1997Be42.
4963.084 *25*	C^{2}S=0.19 for 3/2$^-$ from 1997Be42.
5752.5 *8*	
5841.485 *31*	This level is introduced in S. Raman to P.M. Endt private communication (mentioned in the Adopted Levels of 2011Ch48). Populated by 1144.6γ from 6986 level and deexites by 4268.6γ to 1572 level.
6018.8 *6*	
6078.477 *32*	
6293.93 *6*	
6354.89 *23*	
6419.9 *11*	
6629.43 *9*	
6761.0 *12*	
(6986.099 *24*)	

† From a least-squares fit to γ-ray energies.

^{34}S(n,γ) E=thermal 1985Ra15 (continued)

$\gamma(^{35}S)$

Photons per 100 thermal neutron captures are reported in 1972Dz13 and 1985Ke08. γ-ray cross sections in mb are reported in 1985Ra15, multiplying these by 0.340 yields the number of photons per 100 thermal neutron captures. σ(4638)=163 mb *15*. Iγ(4638)=σ(4638)×0.340=55 *5*.

γ-ray intensities per incident neutron for all transitions and γ-ray cross sections in mb for primary transitions are reported in 1997Be42. The strongest 4638γ is used for normalization. σ(4638)=148.6 mb *55* and Iγ(4638)=σ(1997Be42)/σ(1985Ra15)×Iγ(1985Ra15)=50 *7*.

E_γ [†]	I_γ [†#]	E_i(level)	E_f	Comments
356.66[@] *9*	0.037[@] *4*	2347.782	1991.28	
356.66[@] *9*	0.037[@] *4*	(6986.099)	6629.43	
368.5 *4*	0.020 *7*	2717.05	2347.782	
619.23 *19*	0.042 *8*	3558.082	2938.63	
631.32[@] *24*	0.060[@] *9*	4189.243	3558.082	
631.32[@] *24*	0.060[@] *9*	(6986.099)	6354.89	
^{x}646.9[‡] *5*	0.7 *1*			E_γ,I_γ: from 1972Dz13. 1972Dz13 assigned this γ ray as the 6986 to 6339 transition. This γ ray and its final level 6339 have not been observed in any other ^{34}S(n,γ) measurements. Evaluators consider this γ ray questionable.
^{x}663.41[‡] *7*	0.09 *1*			
692.16 *5*	0.133 *17*	(6986.099)	6293.93	
775.398 *6*	17.2 *19*	2347.782	1572.373	E_γ: other: 775.50 *15* (1972Dz13). I_γ: weighted average of 19.5 *20* (1972Dz13) and 15.6 *17* (1985Ra15).
^{x}803.81[‡] *9*	0.092 *14*			
863.28 *28*	0.034 *10*	3801.955	2938.63	
907.608[‡] *23*	0.56 *10*	(6986.099)	6078.477	E_γ: weighted average of 908.1 *3* (1972Dz13) and 907.607 *14* (1985Ra15). I_γ: weighted average of 0.5 *1* (1972Dz13) and 0.61 *10* (1985Ra15).
1084.79 *15*	0.054 *10*	3801.955	2717.05	
1101.92 *31*	0.033 *8*	4903.374	3801.955	
1144.591 *20*	0.55 *6*	(6986.099)	5841.485	The placement of this γ ray is changed from 2717 to 1572 (1985Ra15) to 6986 to 5841 (S. Raman to P.M. Endt private communication).
1161.05 *20*	0.055 *8*	4963.084	3801.955	
1210.28 *4*	0.252 *27*	3558.082	2347.782	
1250.61 *5*	0.214 *24*	4189.243	2938.63	
^{x}1381.67[‡] *24*	0.024 *8*			
1404.967 *24*	0.60 *6*	4963.084	3558.082	
1454.09 *4*	0.45 *11*	3801.955	2347.782	
1566.7 *3*	0.47 *9*	3558.082	1991.28	
1572.333 *8*	35.0 *34*	1572.373	0	E_γ: weighted average of 1572.15 *16* (1972Dz13) and 1572.333 *7* (1985Ra15). I_γ: others: 39 *4* (1972Dz13) and 32 *4* (1997Be42).
1760.55 *11*	0.146 *21*	4477.62	2717.05	
1840.2 *12*	2.2 *4*	4189.243	2347.782	E_γ: unweighted average of 1839 *1* (1972Dz13) and 1841.426 *15* (1985Ra15). I_γ: weighted average of 3.6 *8* (1972Dz13) and 2.14 *21* (1985Ra15).
1964.8 *2*	0.37 *10*	4903.374	2938.63	
1991.27 *5*	0.54 *6*	1991.28	0	
2022.954 *9*	11 *1*	(6986.099)	4963.084	E_γ: others: 2022.80 *20* (1972Dz13) and 2023.0 *10* (1985Ke08). I_γ: weighted average of 12.0 *15* (1972Dz13), 11 *1* (1985Ke08), 11.4 *10* (1985Ra15), and 9.1 *12* (1997Be42).
2082.83 *17*	15.9 *15*	(6986.099)	4903.374	E_γ: unweighted average of 2082.65 *16* (1972Dz13), 2083.17 *7* (1985Ke08), and 2082.681 *12* (1985Ra15).

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^{34}S(n,γ) E=thermal 1985Ra15 (continued)

γ(^{35}S) (continued)

E_γ [†]	I_γ [†#]	E_i(level)	E_f	Comments
				I_γ: weighted average of 18.0 *15* (1972Dz13), 17 *2* (1985Ke08), 15.6 *17* (1985Ra15), and 12.6 *17* (1997Be42).
^{x}2184.16[‡] *19*	4.7 *5*			E_γ,I_γ: from 1985Ke08. 1985Ke08 assigned this γ ray as a 4903 to 2719 transition. The final level is closest to the known 2717 level. This γ ray has not been observed in any other ^{34}S(n,γ) measurements. Including this γ ray in the level scheme would result in a significant intensity imbalance at its initial level 4903 and its final level 2717 and also lead to poor fitting in the level scheme. Evaluators therefore consider the placement of this γ ray unlikely.
2229.510 *16*	2.86 *27*	3801.955	1572.373	
2347.700 *15*	49 *4*	2347.782	0	E_γ: weighted average of 2347.50 *10* (1972Dz13), 2347.71 *3* (1985Ke08), and 2347.701 *11* (1985Ra15). I_γ: weighted average of 54 *5* (1972Dz13), 49 *4* (1985Ke08), 50 *5* (1985Ra15), and 40 *6* (1997Be42).
2508.39 *8*	0.38 *4*	(6986.099)	4477.62	
2555.492 *14*	3.14 *31*	4903.374	2347.782	E_γ: other: 2555.9 *4* (1972Dz13). I_γ: weighted average of 3.4 *7* (1972Dz13) and 3.09 *31* (1985Ra15).
2615.2 *2*	1.09 *10*	4963.084	2347.782	E_γ: other: 2614.3 *3* doublet (1972Dz13). I_γ: other: <1.3 *4* (1972Dz13).
2616.8 *13*	0.24 *7*	4189.243	1572.373	E_γ: other: 2614.3 *3* doublet (1972Dz13). I_γ: other: <1.3 *4* (1972Dz13).
2716.99 *16*	0.30 *5*	2717.05	0	
2796.74 *4*	4.9 *4*	(6986.099)	4189.243	E_γ: weighted average of 2796.83 *13* (1985Ke08) and 2796.73 *4* (1985Ra15). Other: 2796.6 *5* (1972Dz13). I_γ: weighted average of 6.1 *8* (1972Dz13), 4.3 *4* (1985Ke08), 5.4 *5* (1985Ra15), and 4.9 *7* (1997Be42).
2905.1 *4*	0.14 *4*	4477.62	1572.373	
2938.58 *11*	0.89 *9*	2938.63	0	
2972.0 *4*	0.18 *6*	4963.084	1991.28	
3139.9 *5*	0.078 *24*	6078.477	2938.63	
3183.95 *4*	6.1 *6*	(6986.099)	3801.955	E_γ: weighted average of 3184.30 *20* (1972Dz13), 3183.94 *7* (1985Ke08), and 3183.94 *4* (1985Ra15). I_γ: weighted average of 7.0 *9* (1972Dz13), 5.7 *6* (1985Ke08), 6.2 *6* (1985Ra15), and 5.7 *8* (1997Be42).
3330.84 *13*	7.6 *7*	4903.374	1572.373	E_γ: unweighted average of 3331.08 *15* (1972Dz13), 3330.64 *6* (1985Ke08), and 3330.80 *4* (1985Ra15). I_γ: weighted average of 7.8 *10* (1972Dz13), 7.7 *8* (1985Ke08), and 7.4 *7* (1985Ra15).
3390.62 *7*	5.6 *5*	4963.084	1572.373	E_γ: weighted average of 3390.86 *20* (1972Dz13), 3390.73 *7* (1985Ke08), and 3390.55 *5* (1985Ra15). I_γ: weighted average of 5.5 *8* (1972Dz13), 5.7 *6* (1985Ke08), and 5.5 *5* (1985Ra15).
3558.1 *5*	0.085 *17*	3558.082	0	
3801.73 *4*	2.59 *23*	3801.955	0	E_γ: weighted average of 3802.0 *3* (1972Dz13), 3801.49 *14* (1985Ke08), and 3801.74 *3* (1985Ra15). I_γ: weighted average of 3.6 *7* (1972Dz13), 2.3 *2* (1985Ke08), 3.09 *31* (1985Ra15), and 2.6 *4* (1997Be42).
4105.3 *8*	0.048 *17*	4105.6	0	
4188.95 *5*	2.54 *27*	4189.243	0	E_γ: weighted average of 4188.80 *13* (1985Ke08) and 4188.96 *4* (1985Ra15). Other: 4189.6 *3* (1972Dz13). I_γ: weighted average of 1.9 *4* (1972Dz13), 2.72 *27* (1985Ra15), and 2.8 *4* (1997Be42). Other: 24 *2* (1985Ke08).
4268.65 *14*	0.34 *4*	5841.485	1572.373	E_γ: other: 4268.4 *7* (assigned as 6986 to 2717 in 1972Dz13). The placement of 4268γ is changed from 6986 to 2717 (1985Ra15) to 5841 to 1572 (S. Raman to P.M. Endt, private communication, mentioned in the

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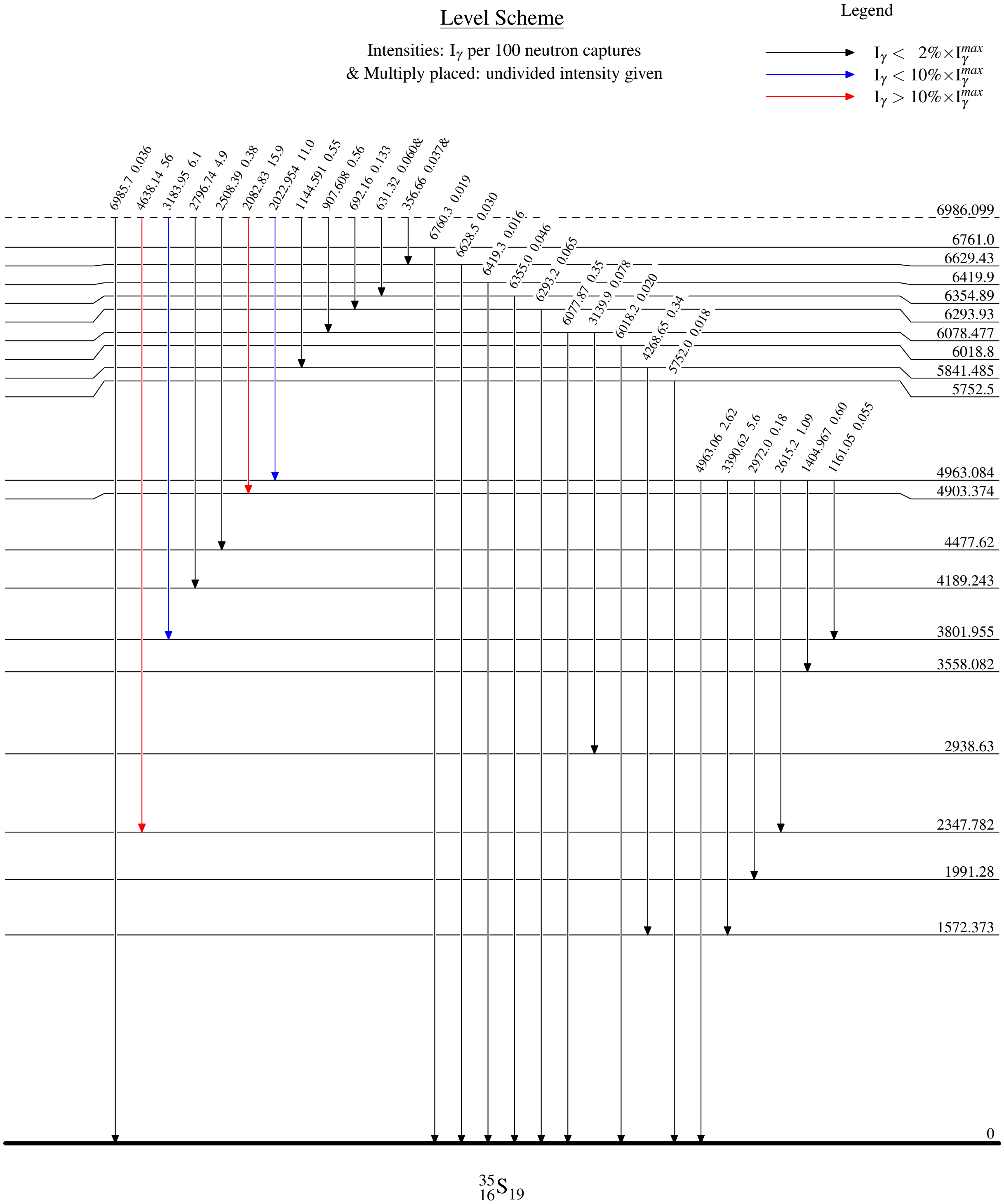
^{34}S(n,γ) E=thermal 1985Ra15 (continued)

γ(^{35}S) (continued)

E_γ †	I_γ †#	E_i(level)	E_f	Comments
				Adopted Levels of 2011Ch48). I_γ: other: 0.4 *2* (1972Dz13).
4638.14 *14*	56 *5*	(6986.099)	2347.782	E_γ: unweighted average of 4638.4 *2* (1972Dz13), 4638.10 *3* (1985Ke08), and 4637.91 *4* (1985Ra15). I_γ: weighted average of 56 *6* (1972Dz13), 59 *5* (1985Ke08), 55 *5* (1985Ra15), and 50 *7* (1997Be42).
4902.99 *5*	4.0 *4*	4903.374	0	E_γ: weighted average of 4903.07 *7* (1985Ke08) and 4902.96 *4* (1985Ra15). Other: 4903.4 *3* (1972Dz13). I_γ: weighted average of 3.2 *8* (1972Dz13), 4.3 *4* (1985Ke08), 3.7 *4* (1985Ra15), and 4.2 *6* (1997Be42).
4963.06 *22*	2.62 *27*	4963.084	0	E_γ: unweighted average of 4963.2 *5* (1972Dz13), 4963.34 *10* (1985Ke08), and 4962.63 *5* (1985Ra15). I_γ: weighted average of 2.6 *8* (1972Dz13), 2.92 *27* (1985Ra15), and 2.18 *33* (1997Be42). Other: 29 *3* (1985Ke08).
5752.0 *8*	0.018 *5*	5752.5	0	
6018.2 *6*	0.020 *7*	6018.8	0	
6077.87 *11*	0.35 *4*	6078.477	0	E_γ: other: 6078.2 *10* (1972Dz13). I_γ: weighted average of 0.3 *1* (1972Dz13) and 0.36 *4* (1985Ra15).
6293.2 *4*	0.065 *14*	6293.93	0	
6355.0 *6*	0.046 *7*	6354.89	0	
6419.3 *11*	0.016 *5*	6419.9	0	
6628.5 *6*	0.030 *6*	6629.43	0	
6760.3 *12*	0.019 *8*	6761.0	0	
6985.7 *10*	0.036 *8*	(6986.099)	0	

† From 1985Ra15, unless otherwise noted.
‡ Not observed in 2007ChZX (PGAA).
# Intensity per 100 neutron captures.
@ Multiply placed with undivided intensity.
x γ ray not placed in level scheme.

$^{34}S(n,\gamma)$ E=thermal 1985Ra15



$^{34}S(n,\gamma)$ E=thermal 1985Ra15

Level Scheme (continued)

Intensities: I_γ per 100 neutron captures
& Multiply placed: undivided intensity given